TASK 5: HEALTHCARE PATIENT ADMISSION TRENDS

Python • Pandas • Seaborn • EDA, length of stay, and IQR outliers

1. Objective and responsible data use

Explore hospital admission counts over time, compare length of stay (LOS) by department, and flag unusual LOS values using the interquartile range (IQR) rule.

- Use only healthcare data you are authorized to access, preferably de-identified.
- Do not include patient names, phone numbers, addresses, or medical record numbers in notebooks, screenshots, charts, or reports.
- An outlier is a record for review, not automatically an error; long stays can be valid.
- Use several years of data when possible before making claims about seasonality.

2. Suggested fields

One row per admission/encounter: pseudonymous admission_id, admission_date, discharge_date, department, age_band, and admission_type. Avoid direct identifiers. Define LOS consistently; a same-day stay may be 0 days if calculated from timestamps.

3. Clean dates and calculate LOS

```
import pandas as pd

df = pd.read_csv("hospital_admissions.csv")
df.columns = df.columns.str.strip().str.lower()
df["admission_date"] = pd.to_datetime(df["admission_date"], errors="coerce")
df["discharge_date"] = pd.to_datetime(df["discharge_date"], errors="coerce")
df["department"] = df["department"].astype("string").str.strip().str.title()

valid = df["discharge_date"].isna() | (
    df["discharge_date"] >= df["admission_date"]
)
df = df[valid].copy()
df["length_of_stay_days"] = (
    df["discharge_date"] - df["admission_date"]
).dt.total_seconds() / 86400
df["month"] = df["admission_date"].dt.to_period("M").astype("string")
print(df.isna().sum())
```

Missing discharge dates may mean that a patient is still admitted, not that the record is invalid. Exclude ongoing admissions from completed-LOS summaries, or report them separately.

4. Monthly admission trend

```
import seaborn as sns
import matplotlib.pyplot as plt

monthly = (df.dropna(subset=["admission_date"])
             .set_index("admission_date")
             .resample("MS")["admission_id"].count())

sns.lineplot(x=monthly.index, y=monthly.values, marker="o")
plt.title("Monthly Hospital Admissions")
plt.xlabel("Month")
plt.ylabel("Admissions")
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig("monthly_admissions.png", dpi=200)
```

```
plt.show()
```

5. Length of stay by department

```
dept_summary = df.groupby("department")["length_of_stay_days"].agg(
    ["count", "mean", "median", "min", "max"]
)
print(dept_summary)

sns.boxplot(data=df, x="department", y="length_of_stay_days")
plt.title("Length of Stay by Department")
plt.xlabel("Department")
plt.ylabel("Length of stay (days)")
plt.xticks(rotation=35)
plt.tight_layout()
plt.savefig("los_by_department.png", dpi=200)
plt.show()
```

Report both mean and median LOS: a small number of very long stays can increase the mean. Include sample counts for each department.

6. IQR outlier detection

Q1 is the 25th percentile and Q3 is the 75th percentile. IQR = Q3 – Q1. Lower fence = Q1 – 1.5 × IQR; upper fence = Q3 + 1.5 × IQR. Values outside these fences are flagged for review.

```
los = df["length_of_stay_days"].dropna()
q1, q3 = los.quantile([0.25, 0.75])
iqr = q3 - q1
lower_fence = q1 - 1.5 * iqr
upper_fence = q3 + 1.5 * iqr

outliers = df[
    df["length_of_stay_days"].notna() &
    ((df["length_of_stay_days"] < lower_fence) |
     (df["length_of_stay_days"] > upper_fence))
]
print("IQR fences:", lower_fence, upper_fence)
print("Flagged records:", len(outliers))
```

7. Interpretation

- Compare the same calendar month across multiple years when assessing seasonality.
- Admission changes may reflect holidays, service capacity, referrals, or data completeness; do not assume a cause.
- Review flagged LOS values against definitions and approved source records; do not delete valid long stays automatically.
- Avoid reporting small groups that could reveal patient identities.

Example wording only: “Monthly admissions peak in July in the supplied sample. Median LOS differs by department. The IQR method flagged some stays for review; the flagged records were not automatically removed.”

Task 5 deliverables

- Approved/de-identified dataset.
- Python notebook or script.
- Monthly admissions line chart.
- Department LOS summary and boxplot.
- IQR calculations, review notes, limitations, and README.